

Supplementary Material: Protect the Decoder

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This supplement holds the tables and figures the main text refers to: the prompt-robustness numbers behind the six-curve figure with their paired tests (Sect. S1), the hook-placement figure (Sect. S2), the held-out DMID results (Sect. S3), per-seed and AUPRC values for the failure detectors (Sect. S4), the 50 px failure-detection table (Sect. S5), the failure-threshold sensitivity check (Sect. S6), the risk-coverage analysis behind the review-budget numbers (Sect. S7), and the calibration of the decoder’s IoU estimate (Sect. S8). All numbers are produced by the released scripts from the per-image result files.

S1 Prompt robustness per expansion level

Table S1 lists the Dice values plotted in the main text’s six-curve figure: LoRA with and without the decoder CKA loss under three training regimes (tight boxes, 20 px jitter, random jitter up to 100 px), evaluated with boxes expanded outward by up to 0, 20, 50, 100 and 200 px. Entries with a standard deviation are means over seeds 0 to 2; the others are seed 0. Table S2 gives the paired Wilcoxon signed-rank test between the two 20 px-trained models at each level, pairs matched on seed and image, with Holm correction over the four datasets at each level.

S2 Hook placement

Figure S1 shows the hook-placement ablation summarised in the main text: LoRA trained with 20 px jitter and no regulariser, with the CKA loss on decoder hooks, on encoder hooks, and on both, evaluated with tight boxes on the two far-ODD sets.

S3 Held-out mammography set (DMID)

DMID was used once, after every model and every hyperparameter was fixed. Table S3 reports tight-box Dice and HD95 on its 269 annotated mammograms for the seven strategies and the CKA-regularised LoRA variants; the ranking matches CBIS-DDSM in every comparison made in the main text. The row marked $n = 2$ is the mean of seeds 1 and 2 of that configuration.

S4 Failure detection: per-seed values and AUPRC

The main text’s detector table averages over three seeds; the seed-1 and seed-2 checkpoints of the five ladder methods were trained for this purpose with the

Table S1. Dice against box expansion for the six models of the main text’s robustness figure. Mean $\pm$ std over three seeds where available.

Model	Dataset	0 px	20 px	50 px	100 px	200 px
no-CKA pm0	ISIC (ID)	0.956 $\pm$ 0.001	0.942 $\pm$ 0.002	0.904 $\pm$ 0.004	0.852 $\pm$ 0.007	0.787 $\pm$ 0.012
no-CKA pm0	PH2 (near-OOD)	0.957 $\pm$ 0.001	0.938 $\pm$ 0.001	0.904 $\pm$ 0.004	0.860 $\pm$ 0.009	0.820 $\pm$ 0.015
no-CKA pm0	BUSI (far-OOD)	0.780 $\pm$ 0.002	0.754 $\pm$ 0.002	0.697 $\pm$ 0.002	0.619 $\pm$ 0.004	0.490 $\pm$ 0.006
no-CKA pm0	CBIS-DDSM (far-OOD)	0.509 $\pm$ 0.010	0.478 $\pm$ 0.015	0.388 $\pm$ 0.013	0.270 $\pm$ 0.014	0.147 $\pm$ 0.004
no-CKA pm20	ISIC (ID)	0.949 $\pm$ 0.002	0.939 $\pm$ 0.001	0.911 $\pm$ 0.004	0.863 $\pm$ 0.009	0.797 $\pm$ 0.015
no-CKA pm20	PH2 (near-OOD)	0.951 $\pm$ 0.001	0.938 $\pm$ 0.003	0.913 $\pm$ 0.006	0.872 $\pm$ 0.009	0.834 $\pm$ 0.008
no-CKA pm20	BUSI (far-OOD)	0.774 $\pm$ 0.032	0.759 $\pm$ 0.034	0.716 $\pm$ 0.035	0.635 $\pm$ 0.030	0.501 $\pm$ 0.017
no-CKA pm20	CBIS-DDSM (far-OOD)	0.437 $\pm$ 0.095	0.419 $\pm$ 0.083	0.358 $\pm$ 0.062	0.247 $\pm$ 0.029	0.133 $\pm$ 0.008
no-CKA rand100	ISIC (ID)	0.941 $\pm$ 0.002	0.937 $\pm$ 0.004	0.926 $\pm$ 0.006	0.902 $\pm$ 0.006	0.848 $\pm$ 0.002
no-CKA rand100	PH2 (near-OOD)	0.950 $\pm$ 0.004	0.944 $\pm$ 0.005	0.933 $\pm$ 0.006	0.910 $\pm$ 0.005	0.877 $\pm$ 0.004
no-CKA rand100	BUSI (far-OOD)	0.747 $\pm$ 0.032	0.740 $\pm$ 0.036	0.716 $\pm$ 0.040	0.654 $\pm$ 0.045	0.496 $\pm$ 0.032
no-CKA rand100	CBIS-DDSM (far-OOD)	0.214 $\pm$ 0.034	0.224 $\pm$ 0.020	0.215 $\pm$ 0.020	0.168 $\pm$ 0.030	0.095 $\pm$ 0.020
CKA pm0	ISIC (ID)	0.953	0.936	0.891	0.822	0.720
CKA pm0	PH2 (near-OOD)	0.952	0.932	0.893	0.830	0.748
CKA pm0	BUSI (far-OOD)	0.871	0.862	0.805	0.702	0.545
CKA pm0	CBIS-DDSM (far-OOD)	0.779	0.714	0.517	0.292	0.146
CKA pm20	ISIC (ID)	0.952 $\pm$ 0.002	0.942 $\pm$ 0.003	0.911 $\pm$ 0.002	0.850 $\pm$ 0.004	0.764 $\pm$ 0.008
CKA pm20	PH2 (near-OOD)	0.954 $\pm$ 0.002	0.942 $\pm$ 0.002	0.912 $\pm$ 0.001	0.861 $\pm$ 0.001	0.800 $\pm$ 0.002
CKA pm20	BUSI (far-OOD)	0.826 $\pm$ 0.008	0.811 $\pm$ 0.013	0.762 $\pm$ 0.019	0.672 $\pm$ 0.023	0.518 $\pm$ 0.023
CKA pm20	CBIS-DDSM (far-OOD)	0.541 $\pm$ 0.095	0.507 $\pm$ 0.077	0.410 $\pm$ 0.059	0.255 $\pm$ 0.031	0.134 $\pm$ 0.014
CKA rand100	ISIC (ID)	0.941	0.937	0.927	0.904	0.851
CKA rand100	PH2 (near-OOD)	0.947	0.943	0.934	0.914	0.880
CKA rand100	BUSI (far-OOD)	0.688	0.675	0.644	0.585	0.466
CKA rand100	CBIS-DDSM (far-OOD)	0.183	0.185	0.164	0.128	0.081

Table S2. Paired per-image Dice difference, CKA minus no-CKA (both 20 px training), at each evaluation level. Improved: fraction of pairs with a positive difference.

Dataset	Level (px)	Pairs	Mean Δ	Median Δ	Improved	p (Holm)
ISIC (ID)	20	3000	+0.003	+0.002	58%	<0.001
ISIC (ID)	50	3000	-0.001	-0.001	49%	0.084
ISIC (ID)	100	3000	-0.013	-0.010	34%	<0.001
ISIC (ID)	200	3000	-0.033	-0.027	24%	<0.001
PH2 (near-OOD)	20	600	+0.004	+0.003	59%	<0.001
PH2 (near-OOD)	50	600	-0.001	-0.000	49%	0.483
PH2 (near-OOD)	100	600	-0.011	-0.009	34%	<0.001
PH2 (near-OOD)	200	600	-0.033	-0.026	17%	<0.001
BUSI (far-OOD)	20	1941	+0.053	+0.012	57%	<0.001
BUSI (far-OOD)	50	1941	+0.046	+0.014	59%	<0.001
BUSI (far-OOD)	100	1941	+0.038	+0.020	61%	<0.001
BUSI (far-OOD)	200	1941	+0.017	+0.010	57%	<0.001
CBIS-DDSM (far-OOD)	20	1086	+0.088	+0.088	64%	<0.001
CBIS-DDSM (far-OOD)	50	1086	+0.052	+0.054	60%	<0.001
CBIS-DDSM (far-OOD)	100	1086	+0.008	+0.015	55%	0.014
CBIS-DDSM (far-OOD)	200	1086	+0.001	+0.004	54%	0.369

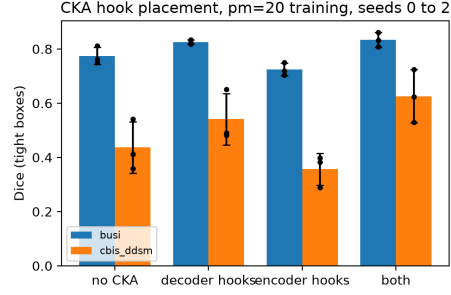


Fig. S1. Far-OOD Dice by hook placement (LoRA, 20 px training, tight-box evaluation). Bars: means over seeds 0 to 2 with one std; dots: individual seeds.

Table S3. DMID, tight boxes, 269 images. HD95 in pixels on the 1024 grid.

Method	Dice HD95 (px)	
Zero-shot	0.637	30
Decoder-only FT	0.727	32
VPT-shallow	0.572	217
VPT-deep	0.526	253
LoRA	0.402	319
Encoder-only LoRA	0.733	25
Full FT	0.727	25
LoRA + CKA (decoder hooks, tight training)	0.697	65
LoRA + CKA (decoder hooks, 20 px) ($n = 2$)	0.556	120
LoRA + CKA (encoder hooks, 20 px)	0.366	330
LoRA + CKA (both, 20 px)	0.651	94

same configurations as seed 0. Table S4 lists the AUROC of each seed with its failure count, and Table S5 the AUPRC as mean $\pm$ std. Table S6 gives AUROC / AUPRC for seed 0 alone, the checkpoints used for the threshold sweep and the risk-coverage analysis.

Table S4. Per-seed AUROC at tight boxes and the number of failures (Dice below 0.5) per seed. Cells with fewer than 20 failures omitted.

Method	Dataset	drift AUROC			IoU-head AUROC			Failures
		s0	s1	s2	s0	s1	s2	
Zero-shot	BUSI (far-OOD)	–	–	–	–	–	–	8/–/–
Zero-shot	CBIS-DDSM (far-OOD)	–	–	–	0.49	–	–	40/–/–
Zero-shot	DMID (held-out)	–	–	–	0.47	–	–	67/–/–
Decoder-only FT	BUSI (far-OOD)	–	–	–	–	–	–	2/3/3
Decoder-only FT	CBIS-DDSM (far-OOD)	–	–	–	–	–	–	12/12/13
Decoder-only FT	DMID (held-out)	0.43	0.26	0.48	0.63	0.49	0.45	53/54/55
VPT-shallow	BUSI (far-OOD)	0.93	0.95	0.96	0.69	0.67	0.64	28/51/60
VPT-shallow	CBIS-DDSM (far-OOD)	0.70	0.75	0.69	0.49	0.44	0.56	78/137/146
VPT-shallow	DMID (held-out)	0.47	0.62	0.63	0.60	0.41	0.62	93/140/140
VPT-deep	BUSI (far-OOD)	0.92	0.91	0.93	0.55	0.68	0.67	44/42/42
VPT-deep	CBIS-DDSM (far-OOD)	0.66	0.71	0.71	0.52	0.42	0.56	113/150/105
VPT-deep	DMID (held-out)	0.45	0.62	0.58	0.53	0.46	0.51	110/143/118
LoRA	BUSI (far-OOD)	0.95	0.96	0.96	0.72	0.40	0.52	61/82/78
LoRA	CBIS-DDSM (far-OOD)	0.82	0.77	0.81	0.46	0.46	0.62	152/130/173
LoRA	DMID (held-out)	0.76	0.56	0.75	0.60	0.40	0.70	175/113/159
Encoder-only LoRA	BUSI (far-OOD)	–	–	–	–	–	–	2/–/–
Encoder-only LoRA	CBIS-DDSM (far-OOD)	–	–	–	–	–	–	12/–/–
Encoder-only LoRA	DMID (held-out)	0.69	–	–	0.24	–	–	52/–/–
Full FT	BUSI (far-OOD)	–	–	–	–	–	–	3/3/3
Full FT	CBIS-DDSM (far-OOD)	–	–	–	–	–	–	14/13/13
Full FT	DMID (held-out)	0.59	0.56	0.57	0.28	0.34	0.33	51/52/50

S5 Failure detection under 50 px box jitter

Table S7 repeats the main text’s detector table with boxes expanded by up to 50 px, the setting in which full fine-tuning and encoder-only LoRA produce enough failures to be scored on the ladder. The IoU estimate detects these prompt-induced failures for every method while drift does not, the pattern discussed in the main text.

S6 Sensitivity to the failure threshold

The main text defines a failure as Dice below 0.5. Tables S8 and S9 recompute the drift AUROC for LoRA and the two VPT variants with the threshold moved from 0.3 to 0.7, at tight boxes and at 50 px jitter. The ordering of methods and

Table S5. AUPRC at tight boxes, mean $\pm$ std over three seeds (single value: seed 0 only).

Method	BUSI (far-OOD)		CBIS-DDSM (far-OOD)		DMID (held-out)	
	drift	IoU head	drift	IoU head	drift	IoU head
Zero-shot	–	–	–	0.12	–	0.27
Decoder-only FT	–	–	–	–	0.20 ± 0.06	0.26 ± 0.08
VPT-shallow	0.68 ± 0.08	0.16 ± 0.01	0.64 ± 0.12	0.35 ± 0.09	0.62 ± 0.17	0.50 ± 0.09
VPT-deep	0.61 ± 0.11	0.14 ± 0.07	0.64 ± 0.06	0.36 ± 0.04	0.59 ± 0.13	0.46 ± 0.04
LoRA	0.80 ± 0.05	0.14 ± 0.04	0.81 ± 0.04	0.45 ± 0.10	0.77 ± 0.15	0.62 ± 0.22
Encoder-only LoRA	–	–	–	–	0.32	0.14
Full FT	–	–	–	–	0.23 ± 0.01	0.13 ± 0.01

Table S6. Seed 0, tight boxes: AUROC / AUPRC for drift and the IoU estimate.

Method	BUSI (far-OOD)		CBIS-DDSM (far-OOD)		DMID (held-out)	
	drift	IoU head	drift	IoU head	drift	IoU head
Zero-shot	–	–	–	0.49 / 0.12	–	0.47 / 0.27
Decoder-only FT	–	–	–	–	0.43 / 0.20	0.63 / 0.35
VPT-shallow	0.93 / 0.58	0.69 / 0.17	0.70 / 0.51	0.49 / 0.26	0.47 / 0.42	0.60 / 0.44
VPT-deep	0.92 / 0.49	0.55 / 0.09	0.66 / 0.57	0.52 / 0.32	0.45 / 0.45	0.53 / 0.42
LoRA	0.95 / 0.75	0.72 / 0.19	0.82 / 0.83	0.46 / 0.41	0.76 / 0.87	0.60 / 0.72
Encoder-only LoRA	–	–	–	–	0.69 / 0.32	0.24 / 0.14
Full FT	–	–	–	–	0.59 / 0.23	0.28 / 0.13

Table S7. AUROC / AUPRC for detecting per-image failures (Dice below 0.5) at 50 px box jitter. Cells with fewer than 20 failures omitted; drift is zero by construction for zero-shot.

Method	BUSI (far-OOD)		CBIS-DDSM (far-OOD)	
	drift	IoU head	drift	IoU head
Zero-shot	–	0.87 / 0.52	–	0.71 / 0.78
Decoder-only FT	0.79 / 0.14	0.48 / 0.05	0.55 / 0.71	0.58 / 0.73
VPT-shallow	0.92 / 0.65	0.49 / 0.15	0.71 / 0.86	0.36 / 0.65
VPT-deep	0.93 / 0.62	0.63 / 0.18	0.61 / 0.81	0.59 / 0.74
LoRA	0.91 / 0.70	0.69 / 0.33	0.76 / 0.91	0.44 / 0.72
Encoder-only LoRA	0.35 / 0.04	0.89 / 0.30	0.36 / 0.54	0.75 / 0.82
Full FT	0.23 / 0.05	0.87 / 0.43	0.32 / 0.51	0.79 / 0.86

datasets is unchanged at every threshold, and LoRA’s values stay within 0.90 to 0.97 on BUSI and 0.73 to 0.82 on CBIS-DDSM at tight boxes.

Table S8. Drift AUROC as a function of the Dice threshold that defines a failure, tight boxes.

Method	Dataset	0.3	0.4	0.5	0.6	0.7
VPT-shallow	BUSI (far-OOD)	0.93	0.94	0.93	0.93	0.88
VPT-shallow	CBIS-DDSM (far-OOD)	0.59	0.65	0.70	0.70	0.71
VPT-deep	BUSI (far-OOD)	0.92	0.90	0.92	0.90	0.84
VPT-deep	CBIS-DDSM (far-OOD)	0.62	0.65	0.66	0.68	0.70
LoRA	BUSI (far-OOD)	0.97	0.94	0.95	0.93	0.90
LoRA	CBIS-DDSM (far-OOD)	0.81	0.82	0.82	0.77	0.73

Table S9. As Table S8, at 50 px box jitter.

Method	Dataset	0.3	0.4	0.5	0.6	0.7
VPT-shallow	BUSI (far-OOD)	0.92	0.91	0.92	0.90	0.88
VPT-shallow	CBIS-DDSM (far-OOD)	0.66	0.68	0.71	0.71	0.65
VPT-deep	BUSI (far-OOD)	0.90	0.91	0.93	0.90	0.87
VPT-deep	CBIS-DDSM (far-OOD)	0.61	0.61	0.61	0.63	0.57
LoRA	BUSI (far-OOD)	0.91	0.91	0.91	0.91	0.90
LoRA	CBIS-DDSM (far-OOD)	0.73	0.75	0.76	0.76	0.71

S7 Risk-coverage analysis

Ranking images by drift and flagging the top fraction for review gives, for each budget, the recall of failures among the flagged images and the failure rate that remains among the unflagged ones (the selective risk). Table S10 lists both at four review budgets, together with the base failure rate and the area under the risk-coverage curve (AURC), for which a random ranking scores the base rate. Figure S2 shows the full curves for drift (solid) and the IoU estimate (dashed).

S8 Calibration of the IoU estimate

Tables S11 and S12 give the ten-bin expected calibration error of the decoder’s IoU estimate $\hat{u}$ against the true IoU on the two far-OOD sets, with the mean of each. After adaptation the estimate sits near 0.5 regardless of the true value; Fig. S3 shows the collapse for LoRA at tight boxes.

Table S10. Tight boxes, failure = Dice below 0.5, ranking by drift. Cells: recall of failures / residual failure rate among unflagged images when the top 10, 20, 30 or 50 percent of images are flagged.

Method	Dataset	Base rate	AURC	top 10%	top 20%	top 30%	top 50%
VPT-shallow	BUSI (far-OOD)	0.043	0.004	0.79 / 0.010	0.86 / 0.008	0.89 / 0.007	1.00 / 0.000
VPT-shallow	CBIS-DDSM (far-OOD)	0.215	0.176	0.29 / 0.169	0.47 / 0.142	0.59 / 0.126	0.73 / 0.116
VPT-deep	BUSI (far-OOD)	0.068	0.009	0.55 / 0.034	0.86 / 0.012	0.91 / 0.009	0.98 / 0.003
VPT-deep	CBIS-DDSM (far-OOD)	0.312	0.265	0.24 / 0.265	0.36 / 0.249	0.50 / 0.221	0.66 / 0.210
LoRA	BUSI (far-OOD)	0.094	0.010	0.70 / 0.031	0.87 / 0.015	0.93 / 0.009	1.00 / 0.000
LoRA	CBIS-DDSM (far-OOD)	0.420	0.258	0.24 / 0.354	0.45 / 0.287	0.61 / 0.233	0.79 / 0.177

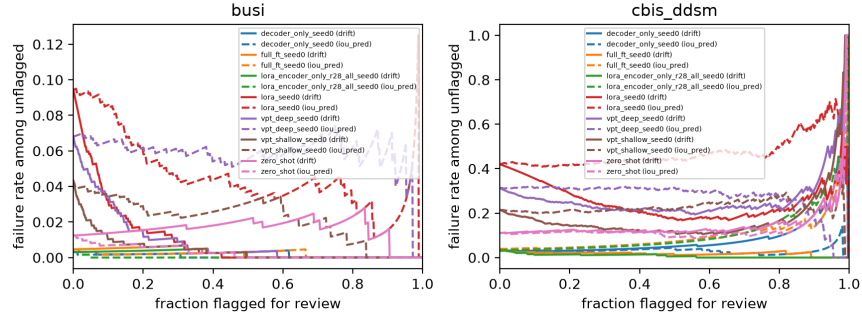


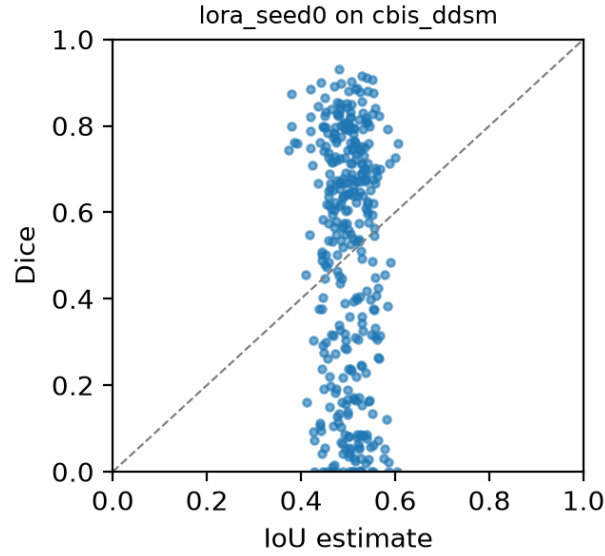
Fig.S2. Residual failure rate among unflagged images against the fraction flagged, tight boxes, for drift (solid) and the negated IoU estimate (dashed). A curve that falls quickly leaves few failures unreviewed.

Table S11. Calibration of the IoU estimate at tight boxes: expected calibration error, mean predicted IoU $\hat{u}$, mean true IoU.

Method	BUSI (far-OOD)			CBIS-DDSM (far-OOD)		
	ECE	$\hat{u}$	IoU	ECE	$\hat{u}$	IoU
Zero-shot	0.16	0.55	0.71	0.09	0.48	0.55
Decoder-only FT	0.34	0.47	0.81	0.24	0.48	0.72
VPT-shallow	0.17	0.53	0.70	0.03	0.53	0.50
VPT-deep	0.17	0.52	0.69	0.07	0.49	0.44
LoRA	0.16	0.51	0.67	0.13	0.50	0.38
Encoder-only LoRA	0.28	0.56	0.84	0.23	0.49	0.71
Full FT	0.26	0.56	0.82	0.22	0.50	0.72

Table S12. As Table S11, at 50px box jitter.

Method	BUSI (far-OOD)			CBIS-DDSM (far-OOD)		
	ECE	$\hat{u}$	IoU	ECE	$\hat{u}$	IoU
Zero-shot	0.07	0.53	0.57	0.20	0.47	0.26
Decoder-only FT	0.18	0.46	0.64	0.20	0.47	0.26
VPT-shallow	0.07	0.53	0.60	0.31	0.54	0.23
VPT-deep	0.09	0.50	0.59	0.28	0.49	0.21
LoRA	0.06	0.51	0.57	0.31	0.50	0.20
Encoder-only LoRA	0.11	0.55	0.67	0.21	0.47	0.27
Full FT	0.10	0.55	0.65	0.21	0.48	0.27

**Fig. S3.** Predicted IoU $\hat{u}$ against Dice per image for LoRA on the far-OOD sets at tight boxes. The estimate stays in a band around 0.5 while Dice spans the full range.